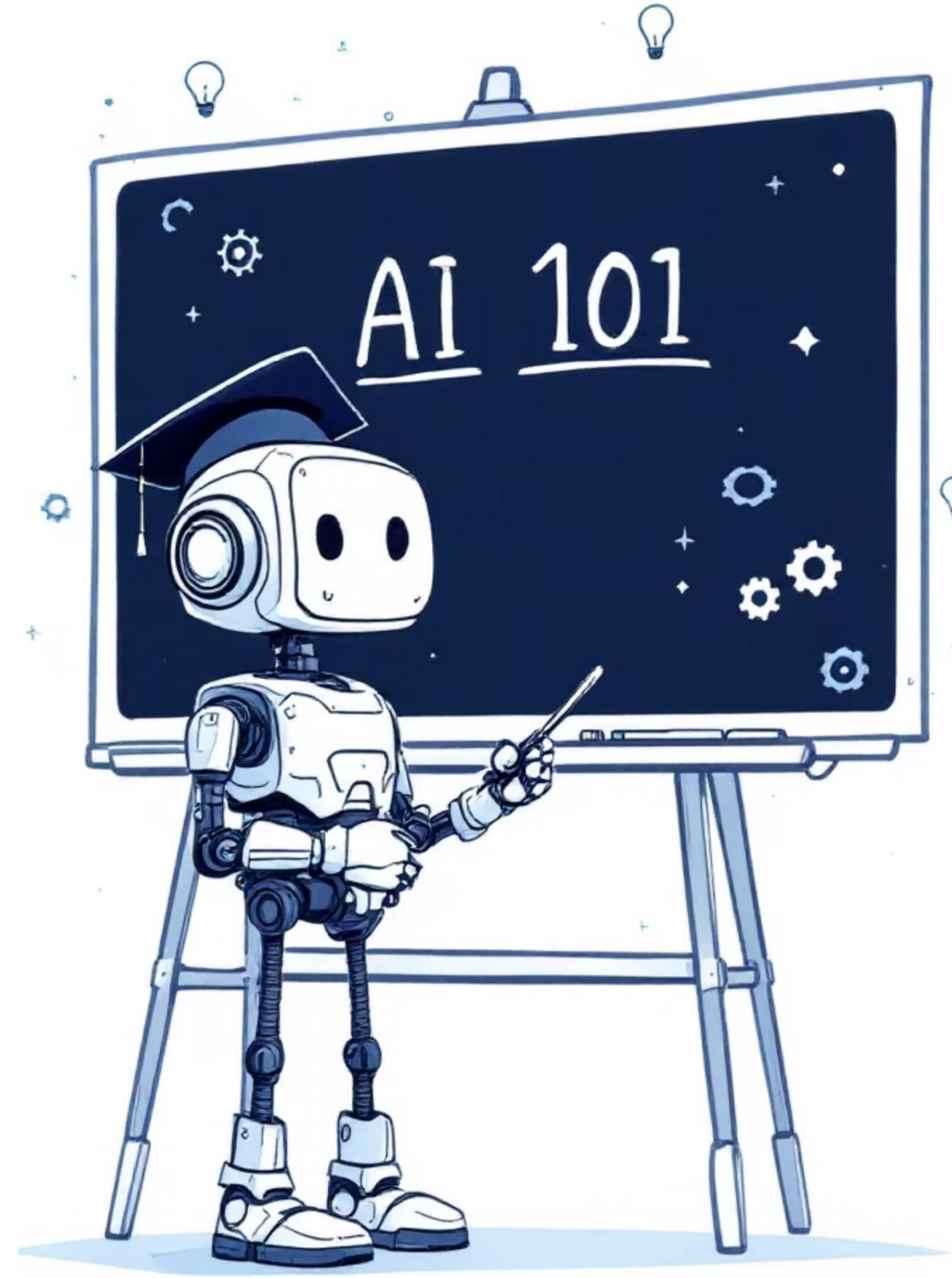


Fundamentals of Artificial Intelligence

AE1FAI — Week 1: Your journey into the world of intelligent machines begins here.



Module Convenors

Dr Qian Zhang

Office: PMB438

qian.zhang@nottingham.edu.cn

Research: Pattern Recognition & Computer Vision —
object recognition, image synthesis, image processing

Dr Xinan Chen

Office:

xinan.chen@nottingham.edu.cn

Research: Optimisation Techniques & Machine
Learning

Today's Session at a Glance

1

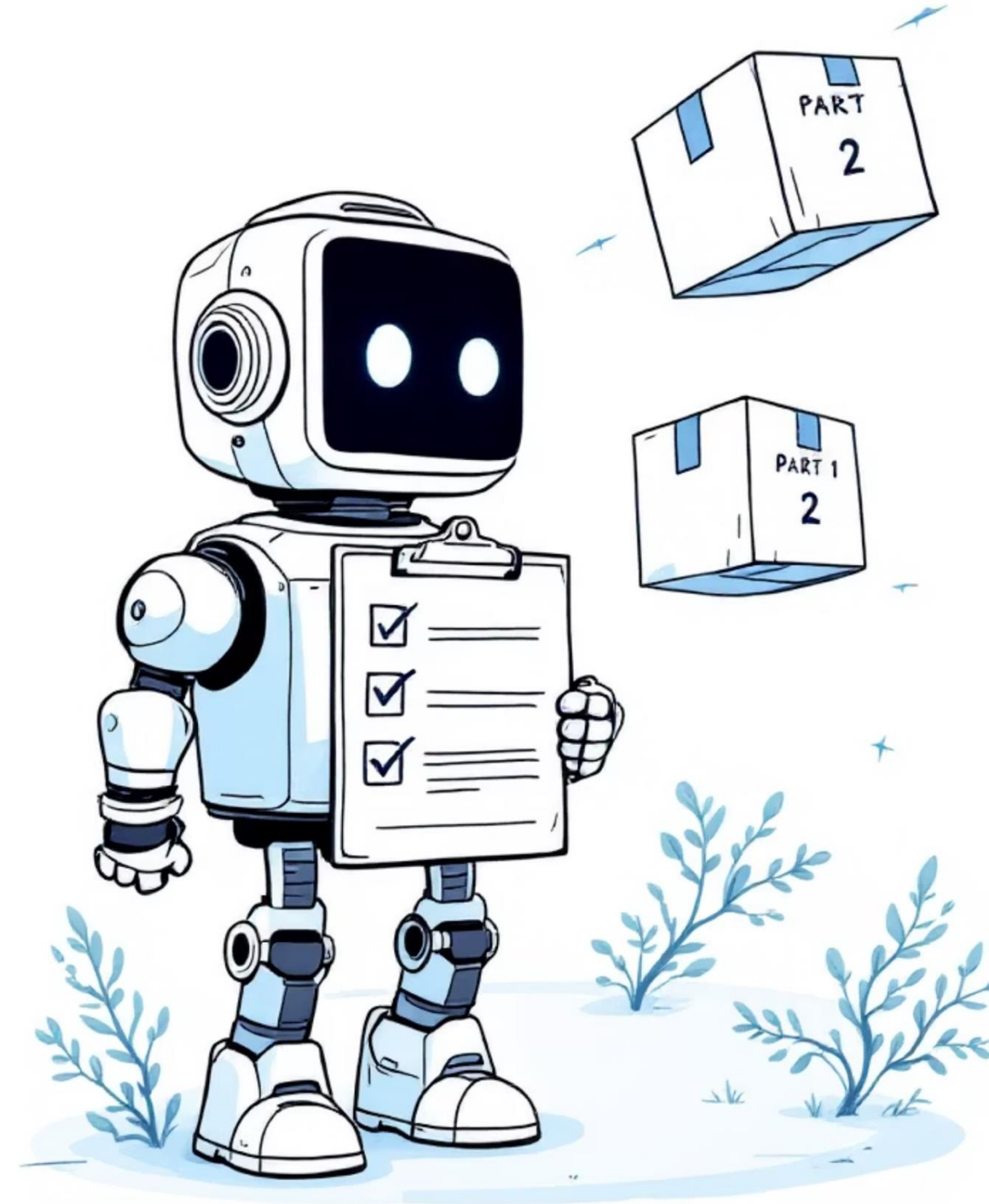
Part 1

Module Introduction — what you'll learn, how you'll be assessed, and the tools you'll use.

2

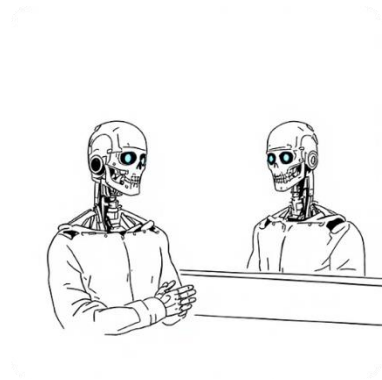
Part 2

What Is Artificial Intelligence? — definitions, history, and the big philosophical questions.



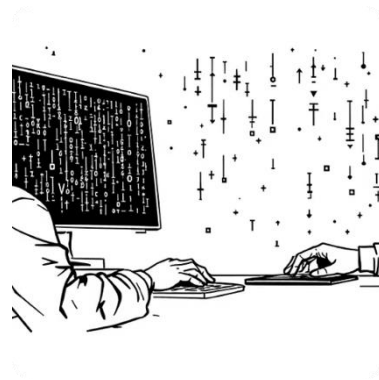
What *Not* to Expect

Let's set some expectations. This module will **not** teach you to...



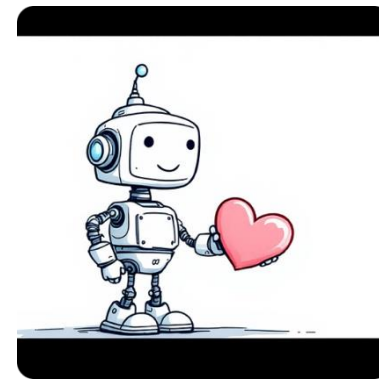
Transform into a Terminator

No cyborg upgrades here, sorry.



Become a Famous Hacker

Hollywood-style hacking is not on the syllabus.



♥ Build a Robot Partner

We'll leave that to science fiction, shall we?

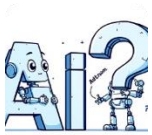


Using ChatGPT or the Latest AI Apps

That's a different course entirely.

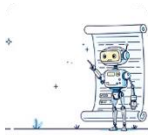
Learning Aims

By the end of this introductory module you will be able to:



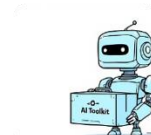
Define AI

Understand what we truly mean by "Artificial Intelligence"



AI Software & History

Get hands-on with AI tools and explore key milestones



Core Techniques

Grasp the basic theory behind a range of AI methods



Applications & Foundations

Discover current AI applications and build a base for future study



Implement

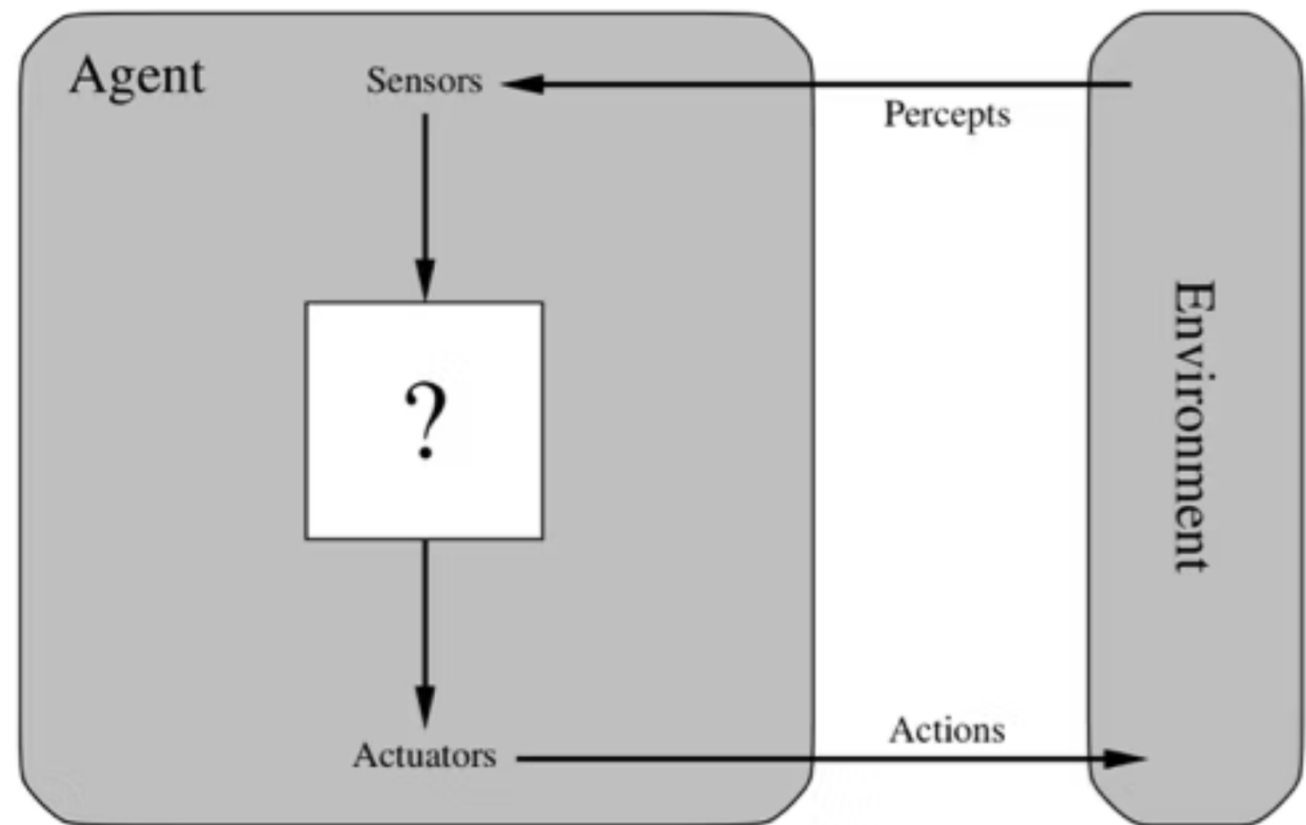
Build and run some fundamental AI techniques in code

Part 1 — What Will You Learn?

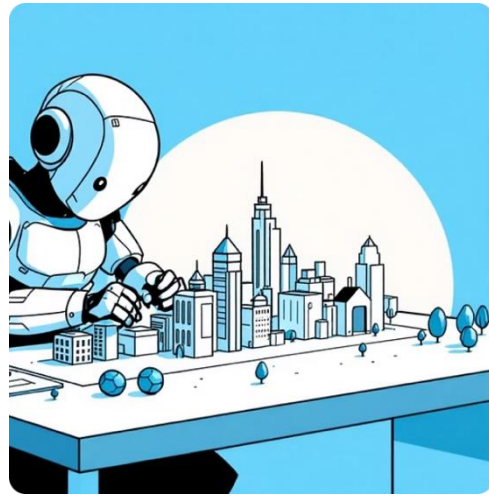
Major AI researchers define this field as:

The study and design of intelligent **agents** — systems that **perceive** their environment and **take actions** that **maximise their chances of success**.

This single definition underpins everything we'll cover this semester.

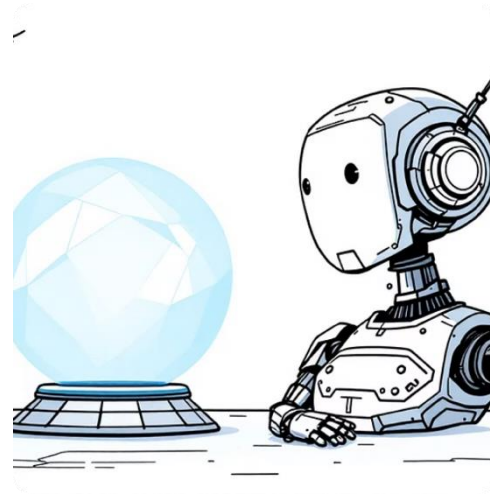


Three Pillars of AI Study



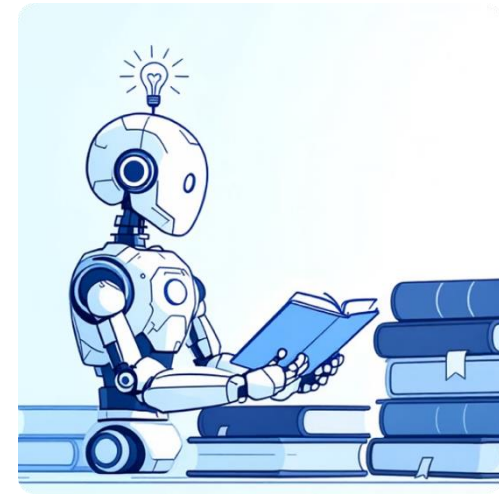
Modelling

Represent the real world in a form a computer can work with.



Inference

Use the model to make **predictions** and draw conclusions.



Learning

Improve the model automatically from data and experience.

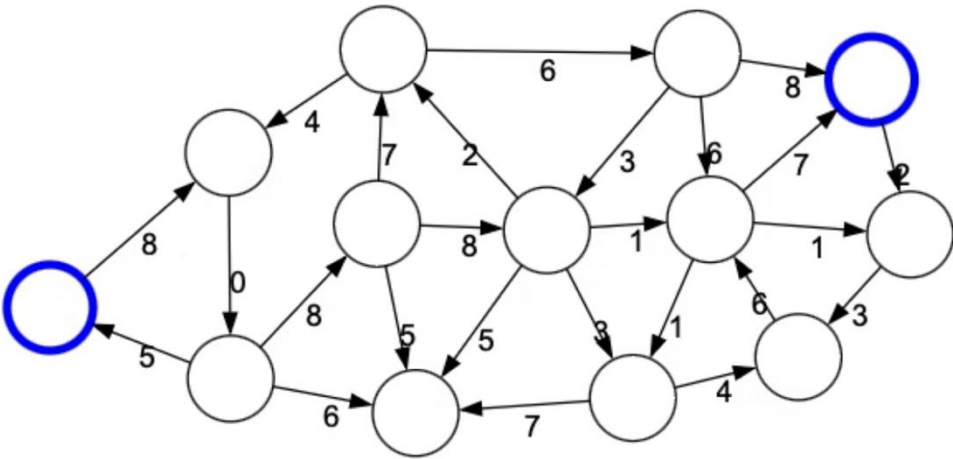
Pillar 1: Modelling

Take a messy real-world problem and turn it into a structured representation a computer can reason about.

Real-World Route Finding



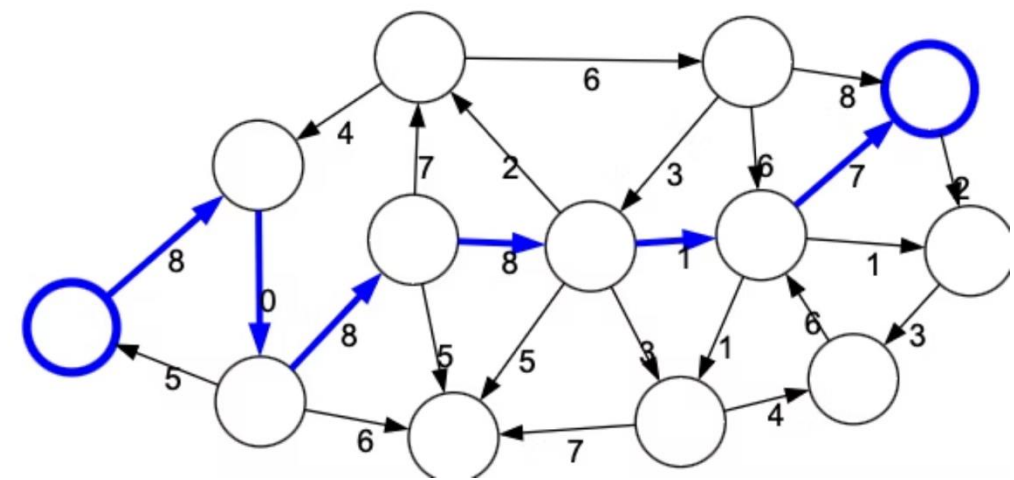
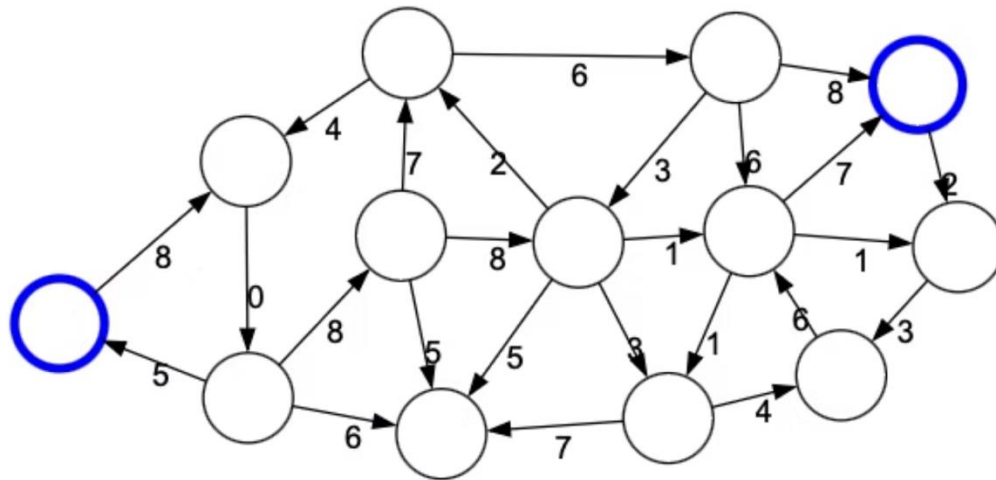
Graph Model



A real road network (left) becomes a simplified graph of nodes and edges (right) — the foundation for search algorithm

Pillar 2: Inference

Once we have a model, we use it to **make predictions**. But what happens when things get tricky?



Can't see the whole graph?

Use heuristic search to guide decisions with estimates.

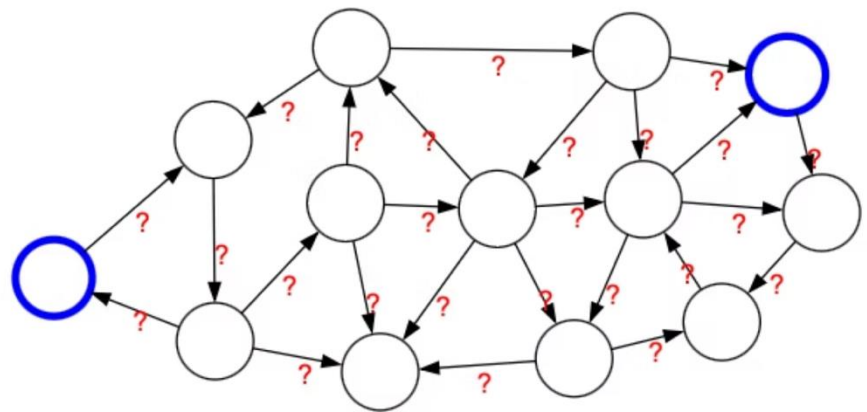
Uncertainty in outcomes?

Use Markov Decision Processes (MDPs) to handle randomness.

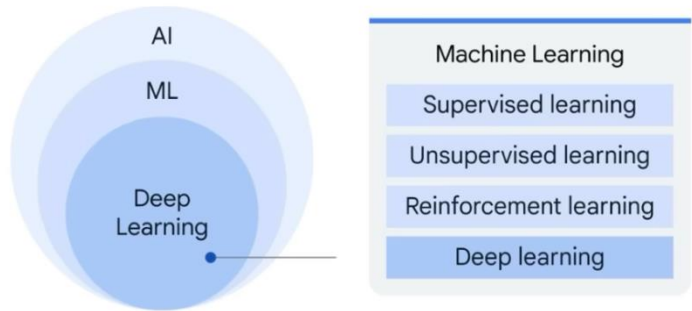
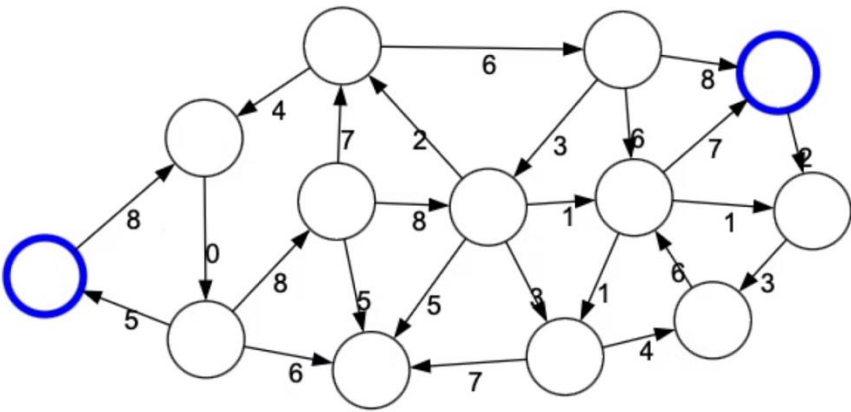
Pillar 3: Learning

Start with a model that has unknown parameters, add **data**, and let the machine discover the right values.

Model without parameters



Model with learned parameters

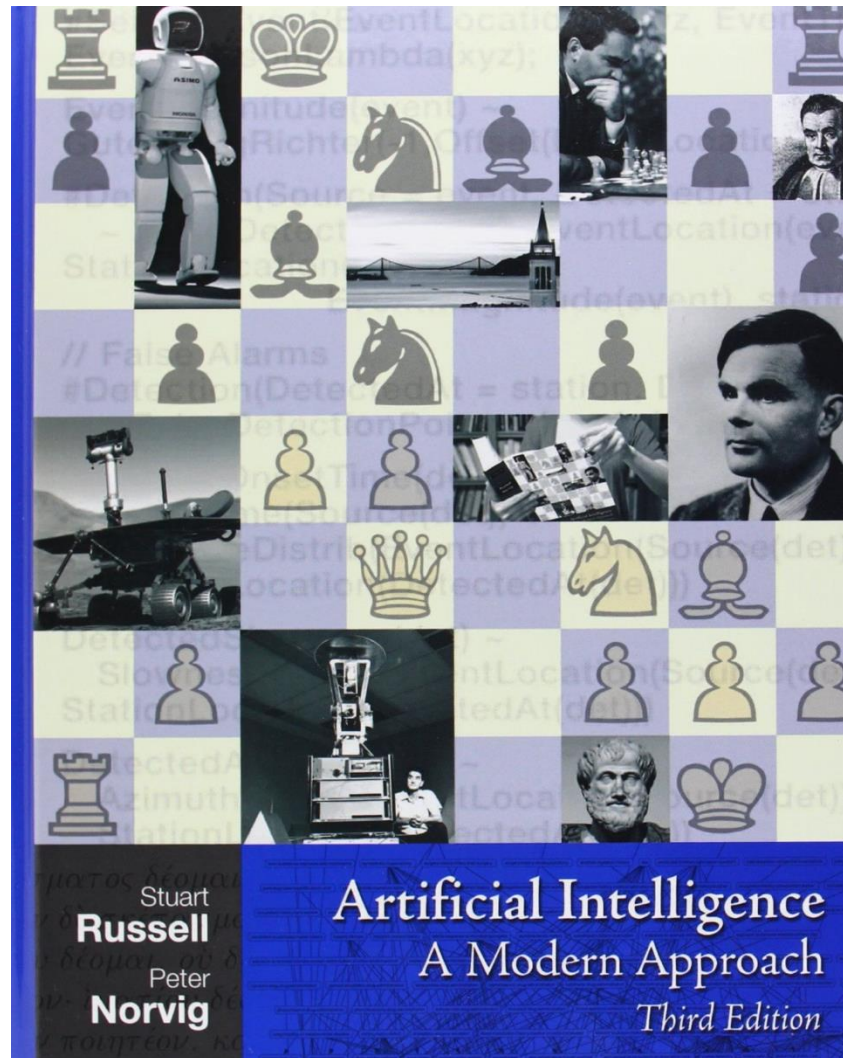


Learning = Model + Data → Tuned Parameters
The machine adjusts itself to fit the patterns hidden in the data.

Module Content Overview

1	Introduction Definition and core issues of AI (Qian) INTRODUCTION
2	Solving Problems by Searching Problem space, Blind search, Game play, Heuristic search (Qian) MODELLING INFERENCE
3	Learning Intro to Data Modelling, Regression and Classification, Neural Networks, Markov Decision Problem (Xinan) LEARNING

Core Textbook



Artificial Intelligence — A Modern Approach

Authors: Stuart Russell & Peter Norvig

Editions: 1995 & 2003 (Global Edition)

- The most comprehensive AI textbook available
- Much of this module's material is drawn from it
- Web: aima.cs.berkeley.edu
- It's a **reference** — you won't need the whole book!

Textbook Chapters We'll Cover

1

Chap 1 — Introduction

2

Chap 3 — Solving
Problems by Search

3

Chap 5 — Adversarial
Search

4

Chap 13.5 — Bayes' Rule

5

Chap 18 — Learning from
Examples

6

Chap 20 — Learning
Probabilistic Models

Plus Chap 26 (Philosophical Foundations) and extra reading provided each session.

Moodle — Your Central Hub

Everything you need lives on the module Moodle page:



Lecture Slides & Notes

All materials uploaded after each session



Assessments

Coursework briefs, exam info, and past papers



Reading List & Resources

Textbook links, schedule updates, and extras



Lab Sessions

Room: IAMET-406 | Language: Python | Start: 5th March (Week 2)

Labs 1–2

Python introduction — allocate extra self-study hours for tutorials

Lab 6

Markov Decision Problems

1

2

3

4

Labs 3–5

Search techniques — blind, heuristic, and game play

Labs 7–8

Machine learning approaches

📄 IDEs: Jupyter Notebook, Spyder, or VS Code — please install **Anaconda** on your laptop before Week 2.

Contact Hours Breakdown



Lectures

20–22 hours across 10–11 sessions



Lab Sessions

16–18 hours of hands-on coding



Coursework

~10 hours of focused assessment work



Self-Study & Revision

~50 hours — the lion's share of your learning



AE1FAI — WEEK 1

Assessment Structure

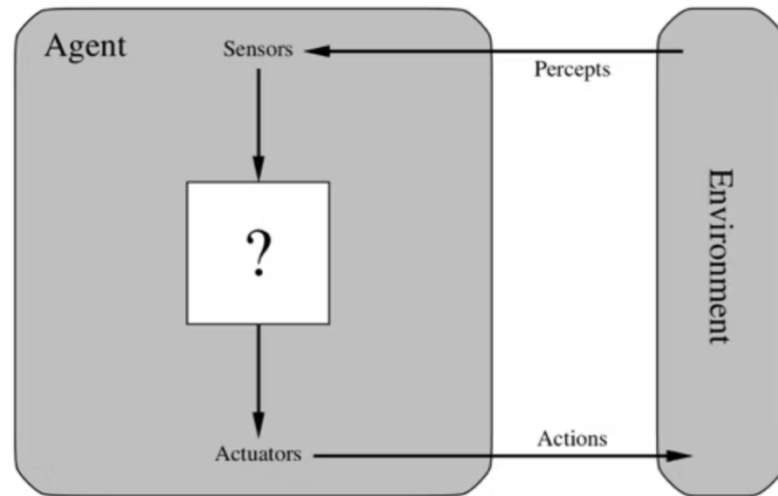
75% — Final Examination

- 2-hour written exam
- 4 compulsory questions
- Covers **all** lecture content

25% — Coursework

- 1 piece of coursework
- Supported in lab sessions

How This Module Fits Your Degree



An intelligent agent must:

- **Perceive** — observations → facts (ML, pattern recognition)
- **Reason** — facts → knowledge (data mining, search)
- **Act** — make decisions (reasoning, optimisation)
- **Maximise success** — optimisation & evolutionary computation

This module is your **foundation**. It feeds into:

COMP2051

AI Methods

COMP2047

Image Processing

COMP3055

Machine Learning

COMP3065

Computer Vision

COMP4107

Big Data

What Is AI?

Consider this: a **five-year-old child** can recognise faces, learn languages, and navigate a room full of obstacles. A **supercomputer** can calculate billions of operations per second.

Which one is more intelligent? And can you think of any real-world examples of AI you've already encountered today?



AI Applications



Planning & Scheduling

Autonomous timetabling and logistics optimisation



Fraud & Spam Detection

Credit card fraud, spam filtering, stock analysis



Computer Vision

Face recognition, medical imaging, quality control



Recommender Systems

Netflix, Spotify, Amazon — personalised for you



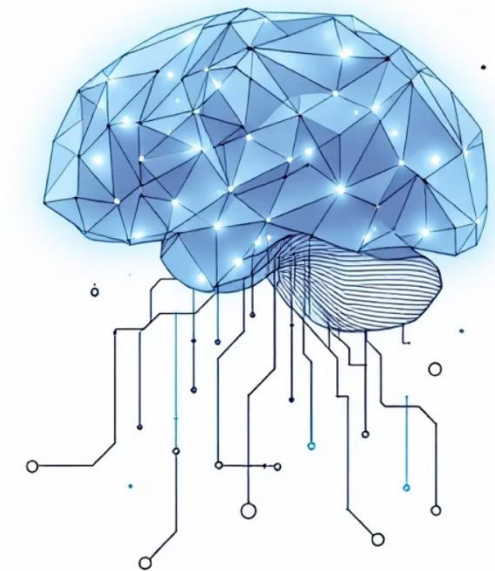
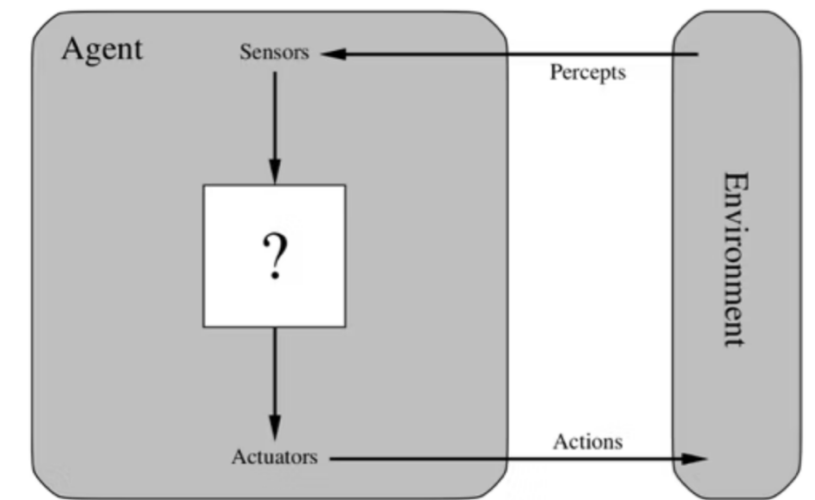
NLP & Speech

Search engines, handwriting and speech recognition

Defining Artificial Intelligence

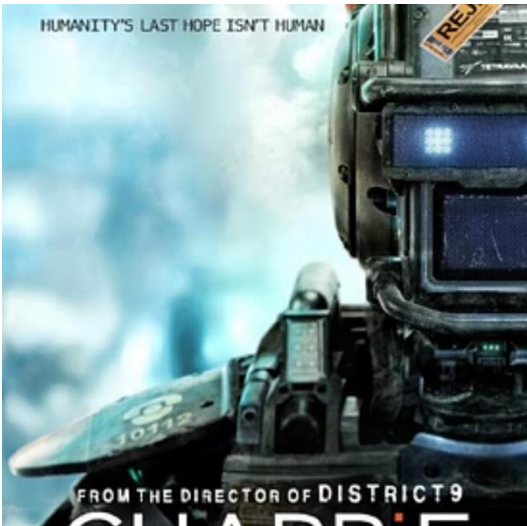
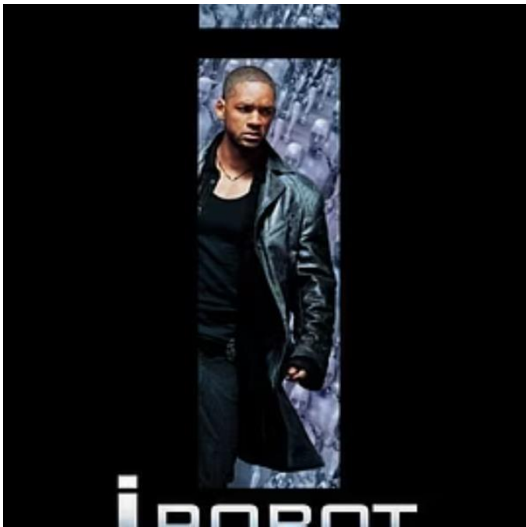
The study and design of **intelligent agents** — systems that **perceive their environment** and take actions that **maximise their chances of success**.

This is the working definition we'll use throughout the module.

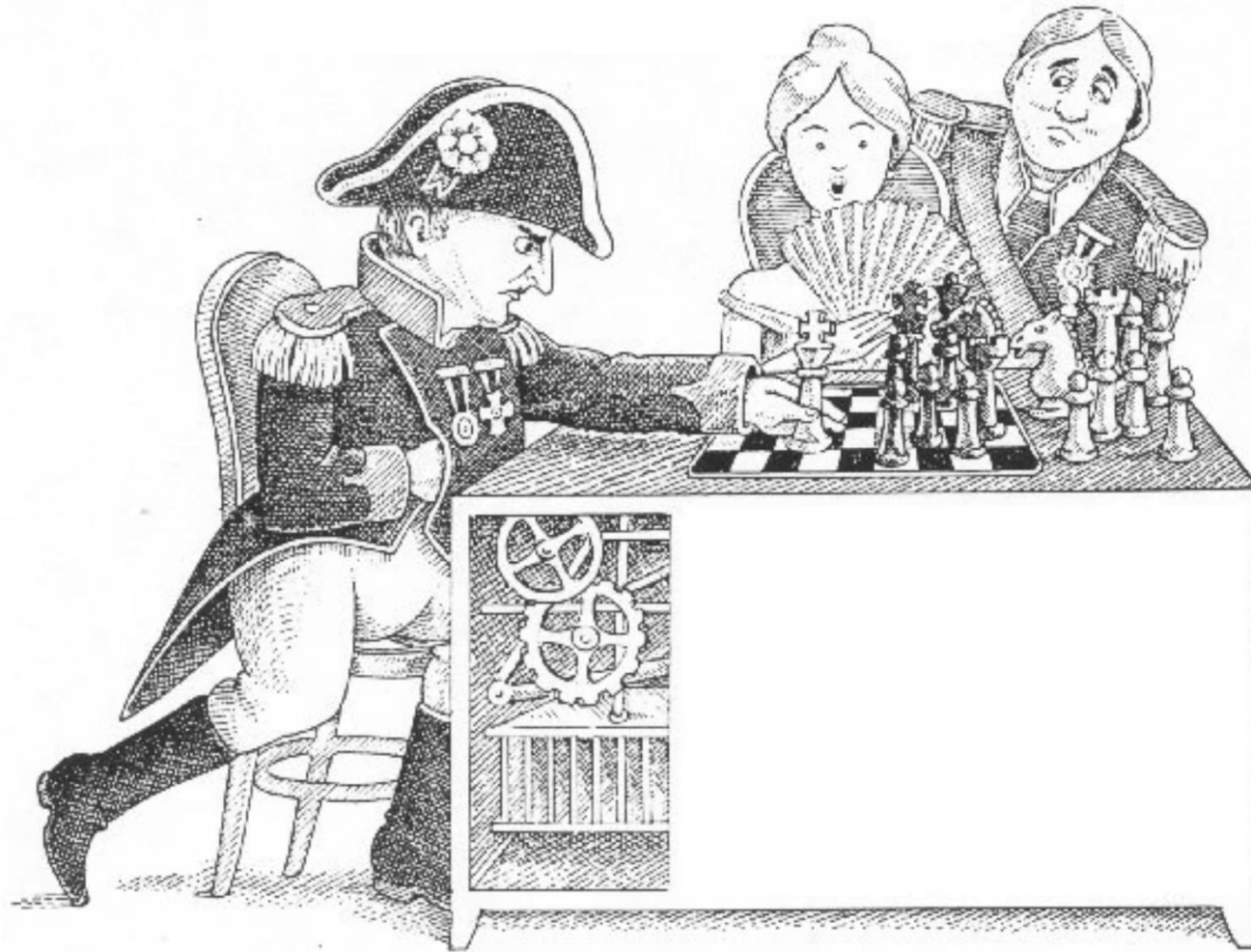


Robots in Movies

The idea that machines could **think** and **act** like humans has inspired countless films and novels.



The Turk — 18th Century Chess Automaton

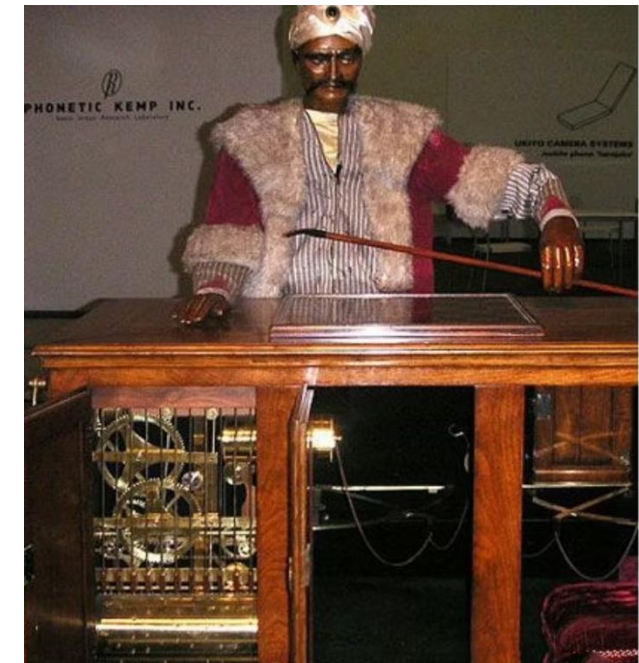


Wolfgang von Kempelen

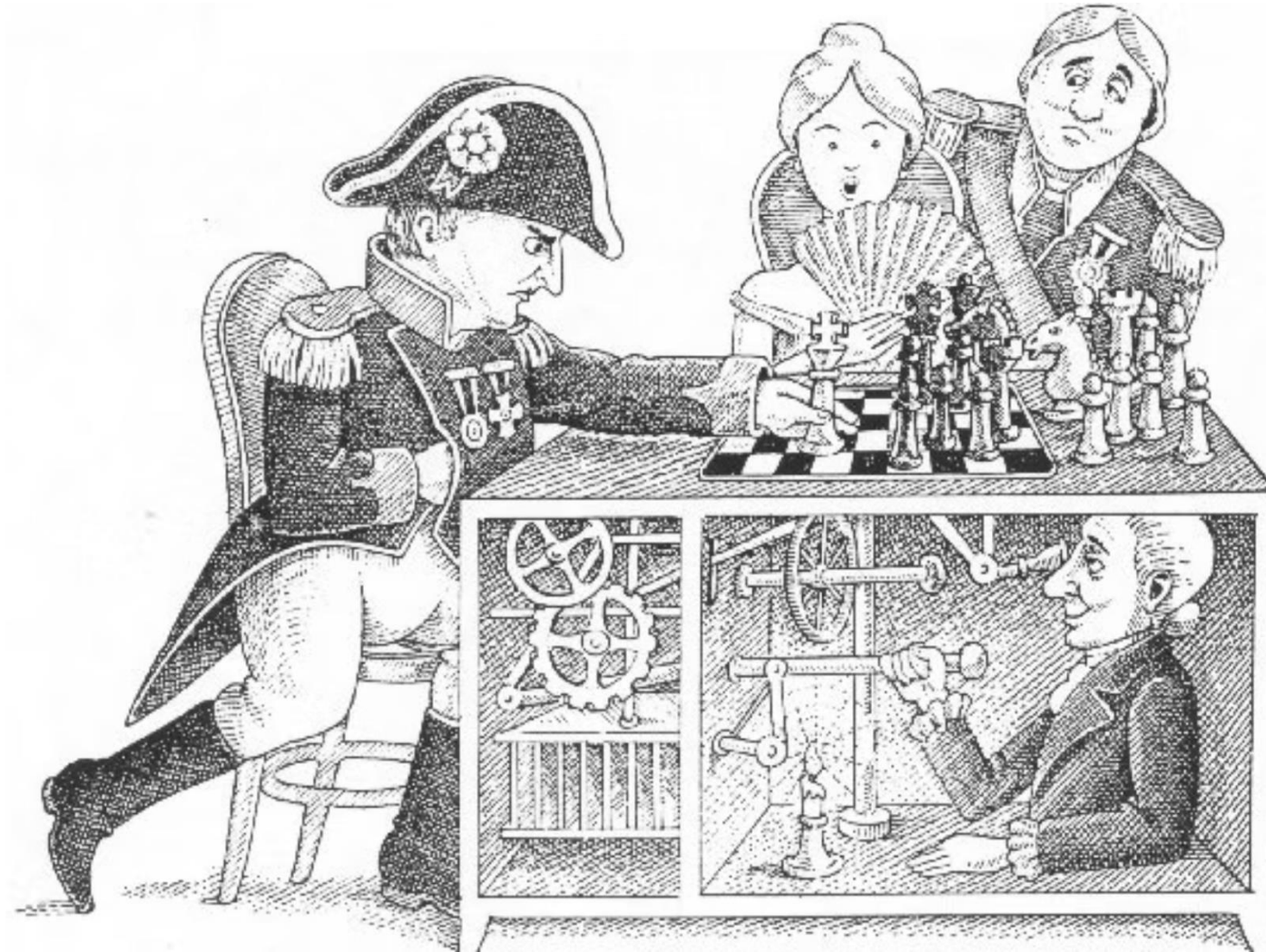
1770–1854

"The Turk" was a mechanical chess-playing automaton that astonished audiences across Europe. It appeared to think independently — but concealed a human chess master inside.

Even as a hoax, it ignited a question that persists to this day: *can a machine truly think?*

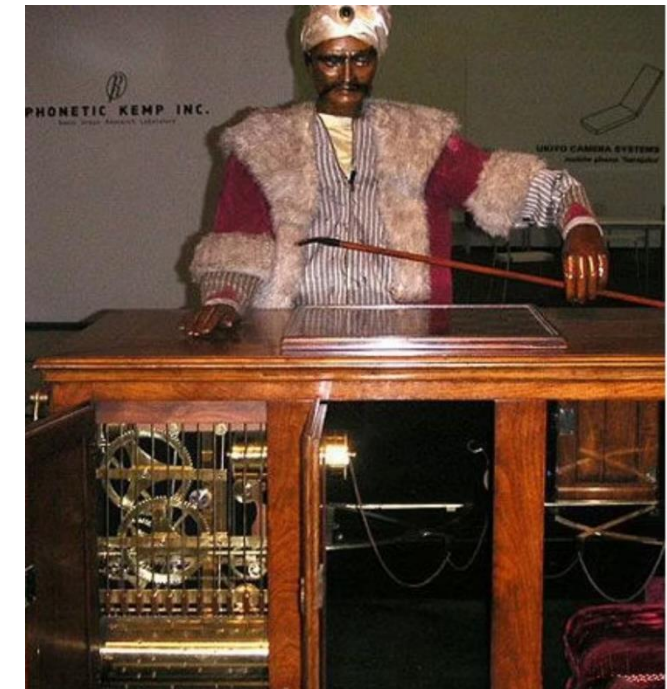


The Secret Inside the Turk



When the cabinet was opened, a hidden **human operator** was revealed, controlling the automaton's moves via an ingenious mechanical system.

The Turk defeated Napoleon Bonaparte and Benjamin Franklin — but the "intelligence" was entirely human. A powerful early lesson: **appearances of intelligence can be deceptive.**



What Is Intelligence?



Understanding Languages

Comprehending meaning, context, and nuance



Automated Reasoning

Drawing logical conclusions from available information



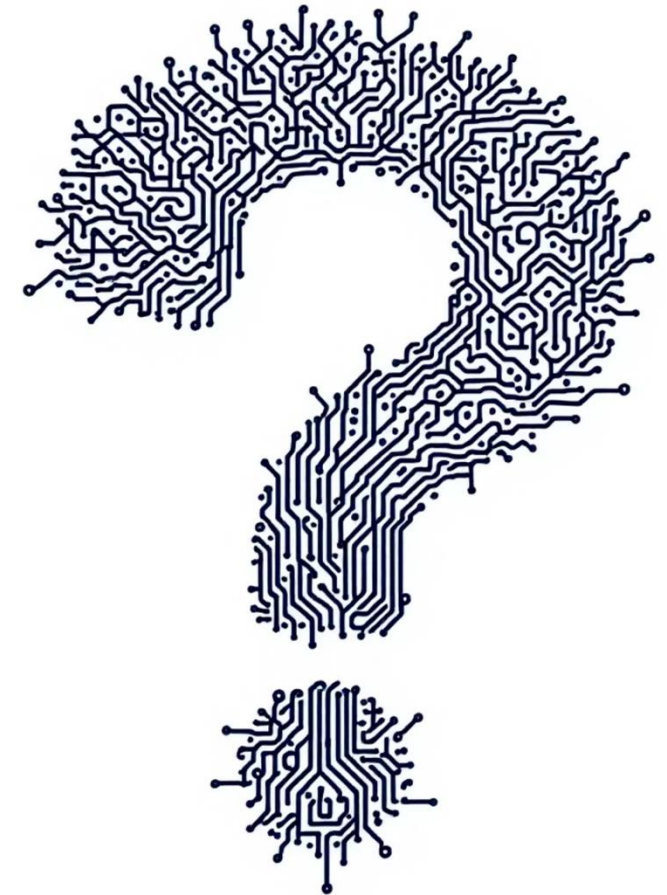
Knowledge

Storing and retrieving facts about the world



Learning

Improving performance through experience



But the deeper question remains: **do machines really understand, or merely simulate understanding?**

Alan Turing — Father of Computer Science



Mathematician, philosopher, and code breaker — Turing's contributions are foundational:

01

1936 — The Turing Machine

Formalised the concept of computation — a model for any general-purpose computer.

02

1939–40 — The Bombe

Designed the machine that cracked the German Enigma cipher in WW2.

03

1950 — The Turing Test

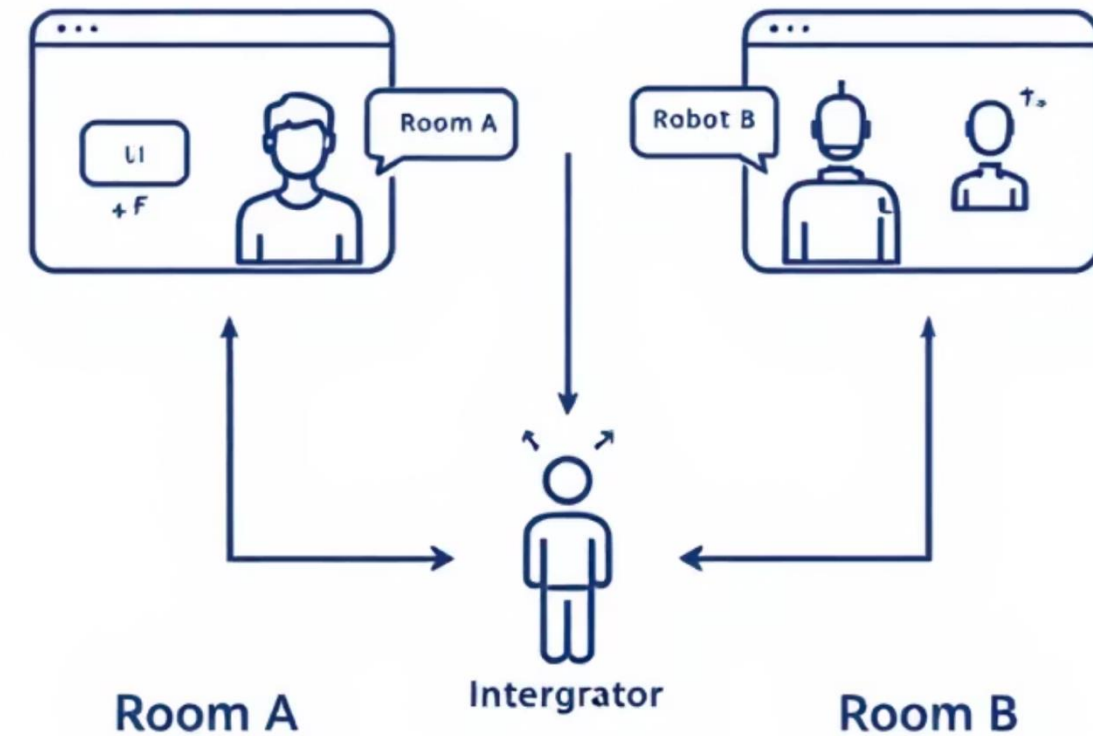
Published his landmark paper on machine intelligence.

The Turing Test — Setup

The test involves **two people and a machine**:

- One person is the **interrogator**, isolated in a separate room
- The other person and the machine are labelled **A** and **B**
- The interrogator **see** either — communication is text-only

The interrogator must determine which respondent is the machine



Ask questions

The interrogator probes both candidates through text-based dialogue only.



Machine tries to fool

The machine's goal is to convince the interrogator it is human.



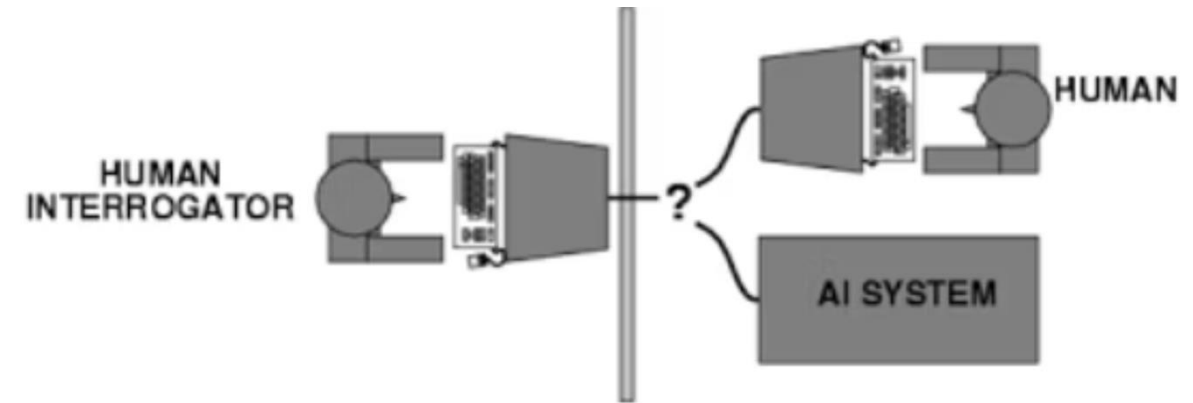
30% threshold

If the machine fools the interrogator **30% of the time**, it is deemed intelligent.

Passing the Turing Test

If the Turing Test is passed, Turing concluded the machine could be considered **intelligent** — or at least capable of **acting intelligently**.

A satisfactory **operational definition** of intelligence: if it behaves indistinguishably from a human, we treat it as intelligent.



Eugene Goostman — First to Pass?

Turing predicted machines would pass the test by **2000**. He was off by 14 years.

In **2014**, the chatbot "**Eugene Goostman**" — posing as a 13-year-old Ukrainian boy — convinced **33% of judges** it was human, becoming the first machine to officially pass a Turing Test.

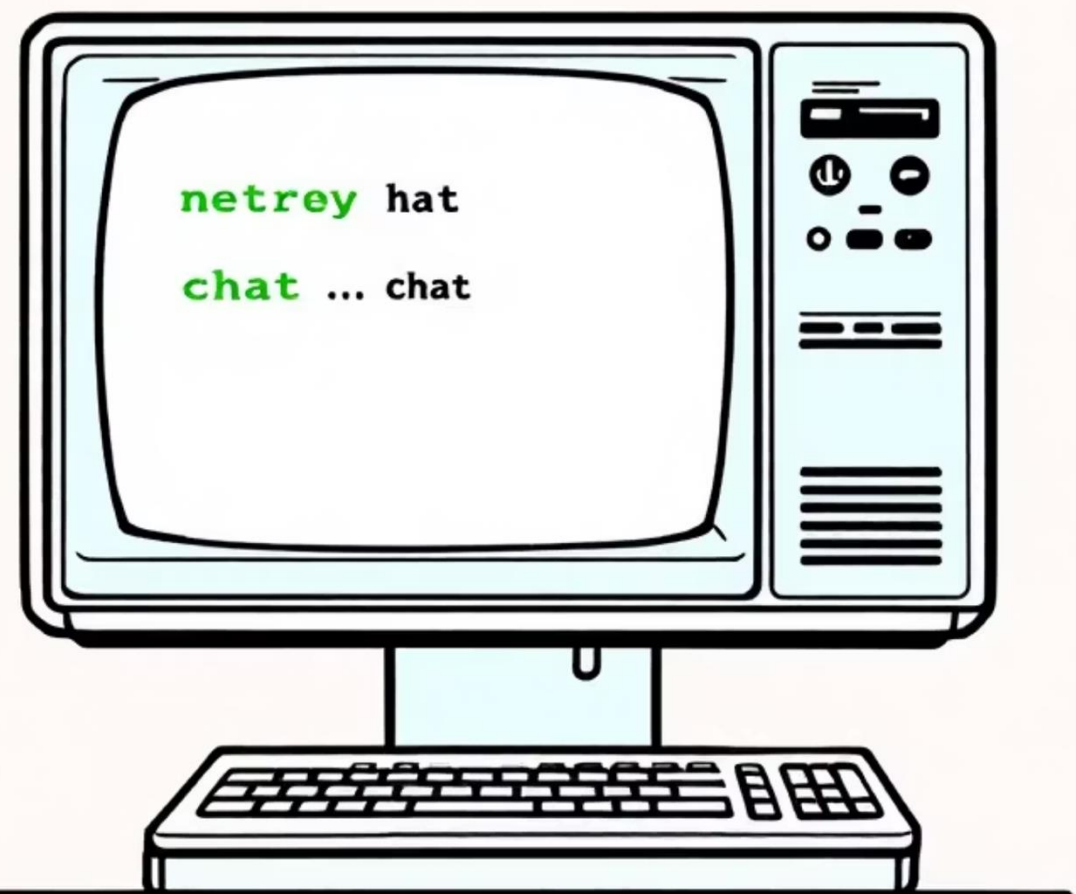
But was it truly intelligent, or just cleverly evasive?

I'LL BE BACK!



EUGENE GOOSTMAN
THE WEIRDEST CREATURE IN THE WORLD

An Interview with Eugene Goostman



Independent

What's it like to be 13, with your life stretching ahead of you?

Eugene

It's something extremely mysterious. I can't tell you anything more about it, sorry... Wanna ask me something more?

Independent

Who's going to win the World Cup?

Eugene

I'm not interested in soccer, unfortunately. I believe even cockroach races are more intelligent and amazing.

Notice how Eugene deflects complex questions with humour and topic changes — a classic chatbot strategy.



AE1FAI — WEEK 1

Try the Turing Test!

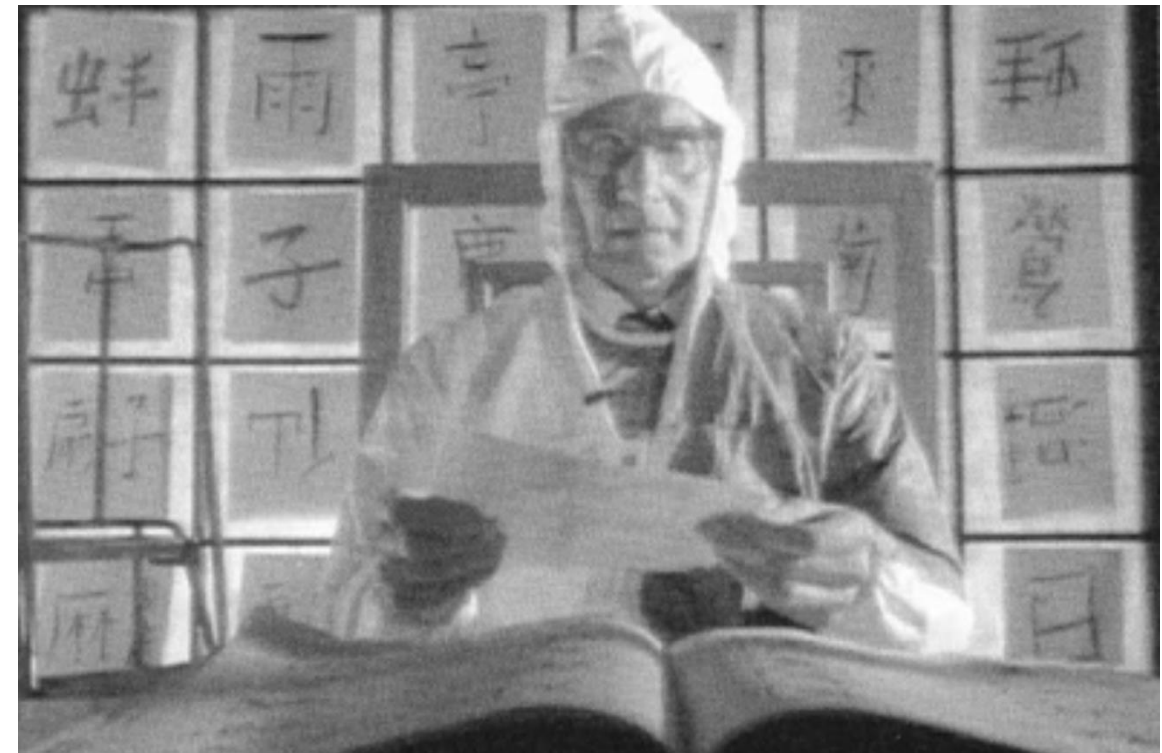
Your turn. Can you tell the difference between a machine and a human through conversation alone? Give it a go and see how convincing modern AI chatbots really are.

The Chinese Room

In 1980, philosopher **John Searle** devised a famous thought experiment to challenge the Turing Test.

Searle argued that **behaving intelligently is not enough** to prove a computer is truly intelligent. *A machine might perfectly simulate understanding without ever actually understanding anything.*

Reference: Searle, J.R. (1980). Minds, Brains and Programs. Behavioral and Brain Sciences, 3: 417–457.



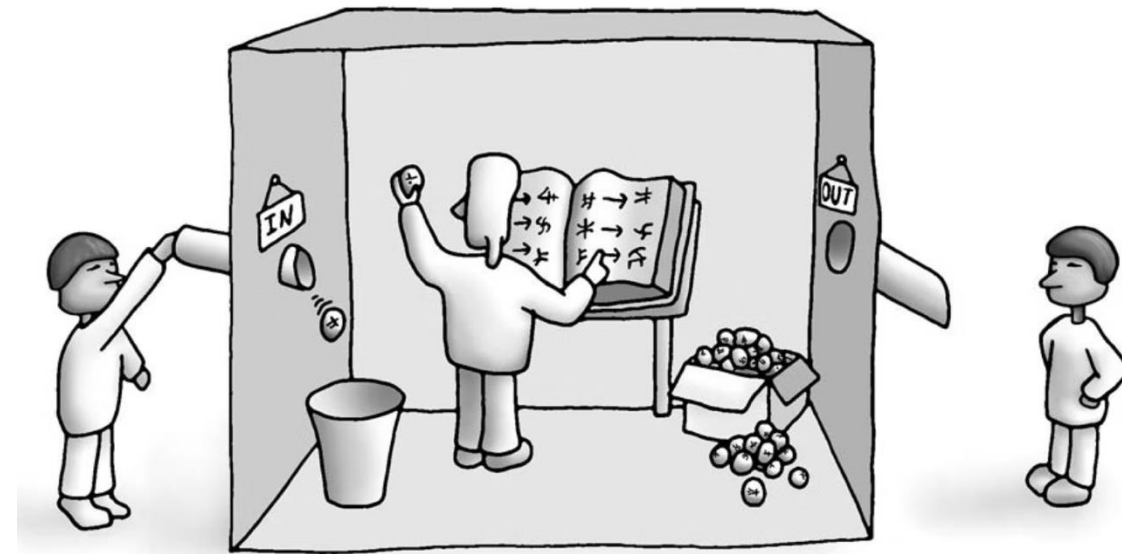
Inside the Chinese Room

The system comprises:

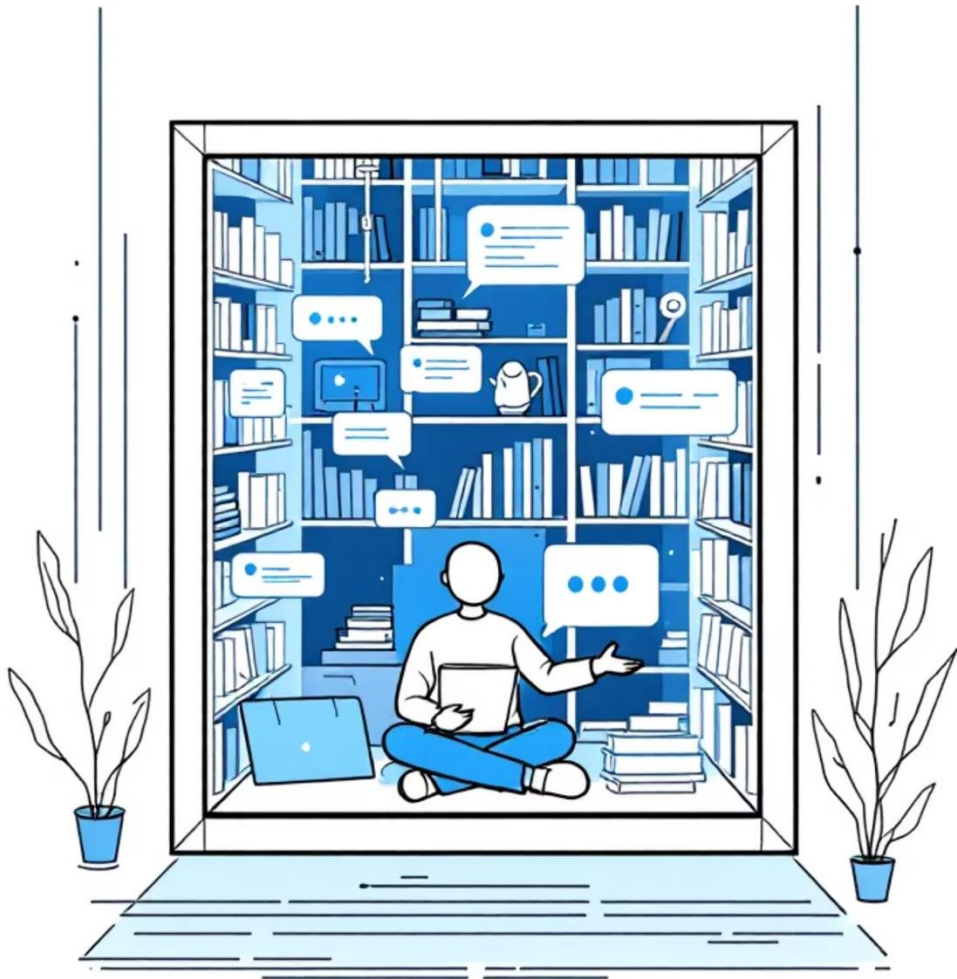
- A **human** who only understands English
- A **rule book** written in English
- Two **stacks of paper** — one blank, one with Chinese symbols

In computing terms:

- The human = **CPU**
- The rule book = **program**
- The paper stacks = **storage**



Searle's Claim



We have a system that **is capable of passing** the Turing Test and is therefore **intelligent** according to Turing.

But the system **does not understand** Chinese as it just comprises a rule book and stacks of paper which do not understand Chinese.

Therefore, **running the right program** does not necessarily generate **understanding**, the system only **act** as it understands Chinese.

Categories of AI

The textbook identifies four perspectives on AI:

	Human	Rational
Think	Thinking humanly	Thinking rationally
Act	Acting humanly	Acting rationally

Strong AI

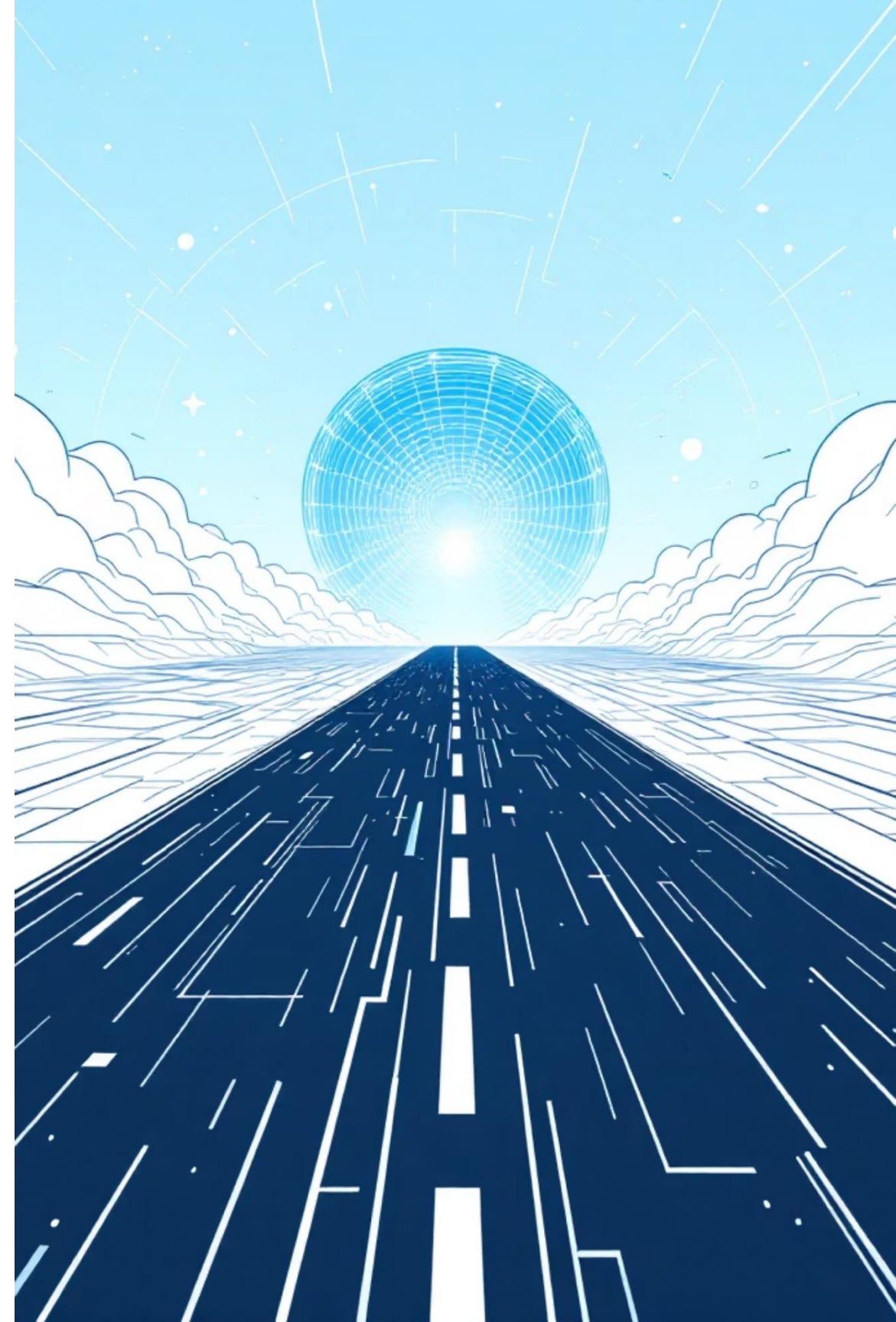
A system can have a **mind** and genuine mental states

Weak AI

A system can **act** intelligently — there are non-intelligent ways to achieve intelligent tasks

The Story of AI

A journey through three ages of artificial intelligence — from reasoning to learning to generation.



Key Moments in AI History

1950 — Turing Test

Alan Turing proposes the **Turing Test**, laying **conceptual foundations** for machine intelligence.

1

2

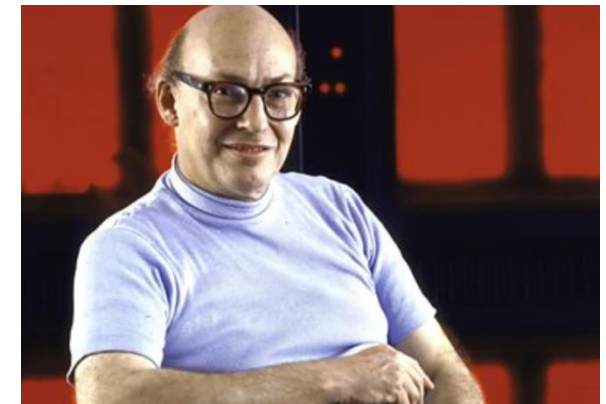
1956 — Birth of AI

John McCarthy and **Marvin Minsky** coin the term "**Artificial Intelligence**" at the Dartmouth Workshop.

Two approaches emerged:

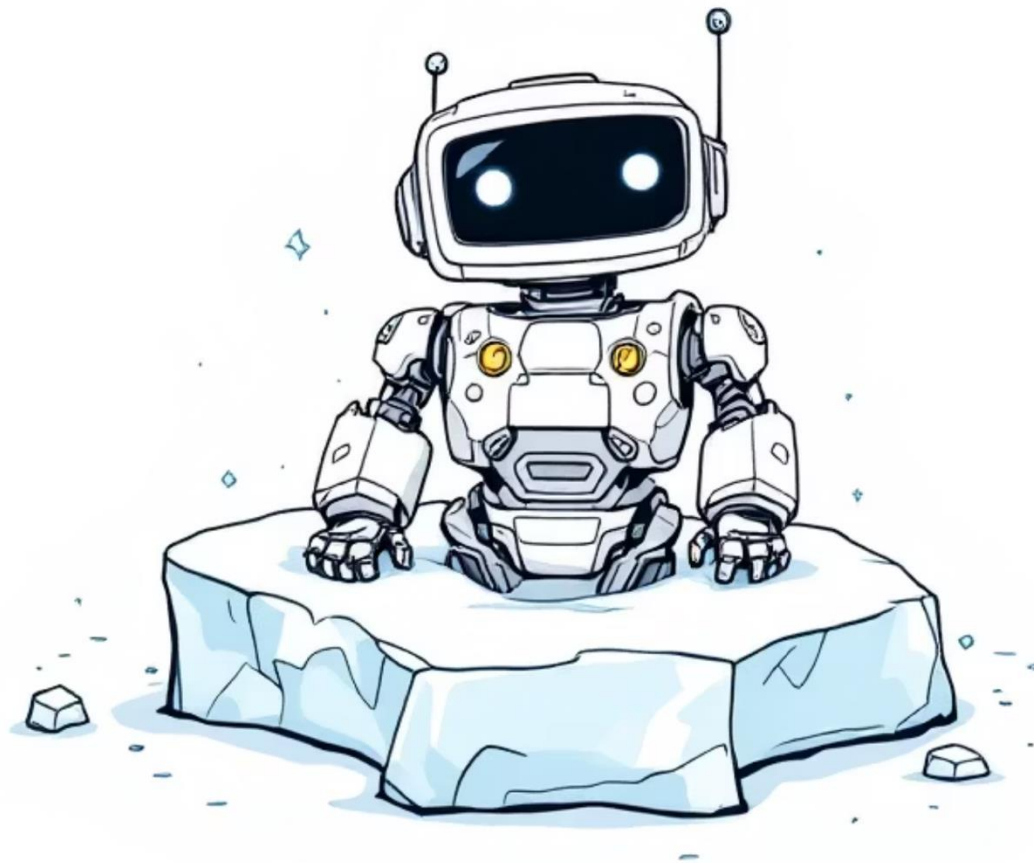
Top-down: pre-programme rules for behaviour

Bottom-up: simulate brains and learn new behaviours



The 1970s AI Winter ❄️

Hopes were high, but reality proved harsh. **Common-sense reasoning** and **simple tasks** like face recognition remained out of reach.



Perceptron Limits



Early neural networks couldn't even implement basic logical functions.

Translation Failure



"**The spirit is willing but the flesh is weak**" became "***The vodka is good but the meat is rotten.***" — true understanding needs context!

Funding Slashed



Governments and industry **lost confidence**. AI research funding collapsed.

The 1980s–2010s: Revival & Milestones

1980s — Expert Systems

Domain-specific "If–Then" rules gave AI **new hope**. General problem-solving gave way to narrow expertise.

1986 — Backpropagation

Geoffrey Hinton rediscovered **backpropagation**, enabling multi-layer neural networks to **learn**.

1989 — CNN for Digits

Yann LeCun used **convolutional neural networks** to recognise handwritten digits.

1997 — Deep Blue

IBM's chess machine **defeats** world champion Garry Kasparov — a watershed AI moment.

2002 — Roomba

iRobot launches the first mainstream **home robot**, bringing AI into living rooms.

2008 — Speech Recognition

Google achieves **92% accuracy** in speech-to-text, making voice interfaces practical.

2012 — AlexNet

A deep neural network achieves massive gains in image recognition — **deep learning** arrives.

2016 — AlphaGo

DeepMind **defeats** world champion Lee Sedol in Go — combining ML with reinforcement learning.

2011 — Watson wins Jeopardy!

IBM's Watson defeats human champions on the **US quiz show**, showcasing NLP prowess.

2014 — Turing Test Passed

Chatbot **Eugene Goostman** fools 33% of judges into believing it's human.

Search & Planning

If we can define the world as a set of **states** and **actions**, intelligence becomes the ability to **search** for a sequence of actions leading to the goal.



Route Planning

Find the shortest path through a road network — GPS, Google Maps.



Logistics & Scheduling

Deliver 100 packages optimally — Amazon, UPS.



Puzzles & Games

Look ahead at possible moves for a winning strategy — Chess, Sudoku.

*This approach dominated AI from the **1960s -1990s**.*

Deep Blue vs. Kasparov (1997)



The Match of the Century

May 1997, New York City. IBM's **Deep Blue** defeated world champion Garry Kasparov **3.5 — 2.5**.

Brute-Force Search

200 million positions calculated per second

Expert Knowledge

Evaluation function **hand-crafted** by Grandmasters — not learned

No Learning

It didn't improve during the game — it just calculated deeper

From Rules to Data

Age 1 Approach

Human programmes rules

"If fever > 38 then flu"

Age 2 Approach

Machine learns rules from data

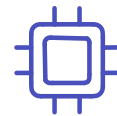
"Data → Pattern → Prediction"

The Three Drivers of the Boom



Big Data

The internet explosion provided massive training datasets (e.g., ImageNet).



Compute Power

GPUs (originally for gaming) proved perfect for parallel neural network maths.



Algorithms

Backpropagation and deep neural networks (Hinton, LeCun, Bengio) matured.

2012 — AlexNet crushed the image recognition competition, proving deep learning works.

Prediction & Personalisation

Instead of following rules, AI finds hidden **patterns** in massive datasets to predict what happens next.



Recommender Systems

"People who liked X also liked Y."
— TikTok, Netflix, Amazon.



Spam Filtering

Detect words like "Free", "Winner",
"Click here" — Gmail, Outlook.



Fraud Detection

Unusual location + high amount +
midnight = alert — bank security
systems.

This is "invisible AI" — working quietly in the background of our daily lives.

AlphaGo vs. Lee Sedol (2016)



The "Unwinnable" Game

March 2016, Seoul. DeepMind's AlphaGo defeated 18-time world champion Lee Sedol **4 — 1** in Go.

Deep Learning (Intuition)

Policy & Value Networks to select and evaluate moves

MCTS (Reasoning)

Monte Carlo Tree Search guided by neural networks

Self-Play

Millions of games against itself to improve

"Move 37 was not a human move. It was beautiful." — Proof that AI can surpass human intuition.

From Classification to Creation

Age 2 — Discriminative AI

"Is this a cat?" Classifying existing data

Age 3 — Generative AI

"Draw me a cat." Creating **new** content

Foundation Models

Instead of **one model per task**, we train **one massive model** on the entire internet (pre-training). This single model adapts to **translation, coding, poetry, and reasoning**.

2017 — Transformer

The architecture

2022 — ChatGPT

Chat interface

2020 — GPT-3

Text generation

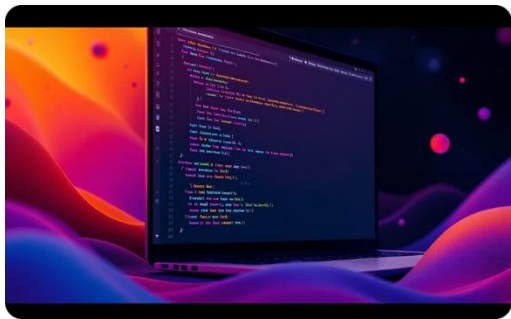
2024 — Sora

Video generation



AIGC Is Everywhere

In 2026, AI is no longer just **analysing** the world — it's **creating** new content across every medium.



Text & Code

ChatGPT, Claude, Copilot — writing essays, debugging Python, summarising emails, translating languages.



Image

Midjourney v6, DALL-E 3 — creating photorealistic art, marketing assets, and UI designs from text prompts.



Video

Sora, Runway Gen-3 — generating high-fidelity video clips, physics simulations, and movie scenes.

PART 5

The Future & What's Next

Autonomous agents, your career, and the ultimate question about machine consciousness.



Mainstream AI Models

The ecosystem has split into two dominant paths: **proprietary giants** and **open challengers**.

Proprietary (Closed)

- GPT-5 (OpenAI) — reasoning & agents
- Claude 4 (Anthropic) — coding & long context
- Gemini 2.5 (Google) — multimodal & real-time

Open Weights / Source

- Llama 4 (Meta) — local deployment & fine-tuning
- DeepSeek R1 (DeepSeek) — efficient reasoning

The Next Frontier: AI Agents



In 2026, AI is moving from answering questions to **taking action** autonomously.

Manus General-purpose agent — plans, browses, codes, and executes complex workflows.	Devin First AI software engineer — builds, debugs, and deploys apps end-to-end.
OpenAI Operator Browser automation agent — navigates websites to book and buy.	Claude Code Agentic coding tool — lives in the terminal, manages git, edits files, runs tests.

Will AI Replace Programmers?



What AI Does Best

- Writing syntax & boilerplate
- Debugging & unit testing
- Translating languages
- Optimising algorithms

"The Hands"



What YOU Do Best

- Defining the problem
- Designing system architecture
- Ensuring ethics & security
- Understanding user needs

"The Head"

"AI will not replace programmers. Programmers who use AI will replace those who don't."

Predictions & The Ultimate Question

"Predicting the future of AI has historically been... difficult."

Herbert Simon (1957)

"Within ten years a digital computer will be the world's chess champion."

Reality: Took 40 years.

Alan Turing (1950)

"One will be able to speak of machines thinking without expecting to be contradicted."

Reality: Mostly true.

Ray Kurzweil

"The Singularity will happen by 2045. AI will surpass all human intelligence combined."

Stephen Hawking

"The development of full AI could spell the end of the human race."

The Hard Problem: Can a machine ever truly *feel*, or will it always just be simulating?

Week 1 Summary

Here's what we covered today — make sure you're comfortable with each point before next week.



Four Categories of AI

Thinking/acting × humanly/rationally



Strong vs. Weak AI

Genuine understanding vs. intelligent behaviour



Turing Test & Chinese Room

Two landmark thought experiments on machine intelligence



AI History & Milestones

From the Dartmouth Workshop to AlphaGo and beyond

